

1. Machine Learning in the Payment Industry

The payment industry processes billions of transactions every day, and the ability to make fast, accurate decisions is critical. Traditional rule-based systems are no longer sufficient to handle the complexity and scale of modern payment flows. This is where Machine Learning (ML) comes in.

ML allows payment companies to analyze large volumes of data, identify patterns, and make predictions in real time – without manual intervention. It helps businesses reduce fraud, improve payment success rates, and better understand customer behavior.

In this document, we explore three key areas where ML is actively used in payments, look at real-world examples of companies applying it, and discuss the main challenges that come with implementation.

Key Directions of ML Application

1. Fraud Detection

Fraud is one of the biggest challenges in the payment industry. Every year, billions of dollars are lost due to unauthorized transactions, stolen card data, and account takeovers. Traditional rule-based systems (for example, "block all transactions over \$1000 from a new device") are too rigid – they either miss real fraud or block legitimate payments.

ML solves this by analyzing hundreds of signals at once – transaction amount, location, device, user behavior history, time of day – and calculating a fraud probability score in real time. If the score is too high, the transaction is blocked or sent for additional verification.

Example: Stripe

Stripe built a fraud prevention tool called Radar. It assigns a risk score to every payment using hundreds of signals – card data, IP address, device information, and transaction history. The model is trained on data from millions of businesses processing more than \$1.9 trillion in payments per year, meaning there is a 92% chance any given card has already been seen on the network before. As a result, Radar reduces fraud by 32% on average. (URL: <https://stripe.com/radar>) (URL: [ByteByteGo Newsletter](#))

2. Payment Authorization Optimization

Even legitimate payments are often declined by banks – due to fraud suspicion, technical errors, or outdated rules. This directly reduces business revenue. ML helps predict which routing, timing, or request parameters give the highest chance of a successful authorization.

Example: Adyen

Adyen launched an AI product suite called Uplift, which uses ML to automate decisions about transaction routing and optimization, drawing on data from more than one trillion dollars worth of processed payments. In practice, it increases authorization rates by 6% on average while cutting payment costs by 5% – with merchants like eBay and Microsoft already seeing results. (URL: [Adyen Launches AI Payment Suite with 6% Conversion Boost](#)) (URL: [How Adyen's new AI platform boosts conversion while reducing fraud](#))

3. Risk Scoring & Chargeback Prevention

Chargebacks – when a customer disputes a transaction and the bank reverses the payment – cost businesses not just the transaction amount but also additional fees and penalties. ML helps predict which transactions are likely to result in a chargeback before it happens, allowing businesses to take action early.

Example: Mercari

Mercari, a Japanese e-commerce marketplace, faced a problem: their fraud prevention team could not manually review every transaction due to volume. They built an ML system called CBML – a binary classification model that detects potential chargebacks among normal transactions. The model found a key pattern in the data: if one chargeback happens from a buyer, later transactions from that same buyer have a high probability of also being chargebacks. (URL: <https://engineering.mercari.com/en/blog/entry/chargeback-ml/>)

Challenges

Data Quality ML models are only as good as the data they are trained on. In payments, data is often incomplete, inconsistent, or imbalanced — for example,

fraud cases make up less than 1% of all transactions, which makes it hard for a model to learn what fraud looks like.

Model Scaling A model that works well for 1 million transactions per day may fail at 1 billion. Payment systems require decisions in milliseconds at massive scale, which puts significant pressure on infrastructure.

Regulatory Constraints Payment companies operate under strict regulations such as GDPR and PSD2. ML models must be explainable – regulators often require that a business can explain why a transaction was declined, which is difficult with complex deep learning models.

Cold Start Problem When a new merchant or card joins the network, there is no historical data to base decisions on. The model has no context and is more likely to make errors – either blocking legitimate payments or missing fraud.